

Chapter 5: Enterprise Metamodel

Overview

Chapter 5 (pp. 63-110) of **Mastering the TOGAF® Standard** by Eric Jager (2025) explains the **Enterprise Metamodel**, one of the core components of the TOGAF Standard and the conceptual root of every entity used in architecture work described in the (Extended) Content Framework. The chapter is part of the GREEN, BLUE, and GOLD reading tours (p. 63). It walks through all 36 entities of the Foundation Enterprise Metamodel, organized into five interrelated sections (General Entities, Business Architecture, Data Architecture, Application Architecture, Technology Architecture), using a running example of a fictional healthcare organization. It closes by explaining how the metamodel must be tailored to an organization during the ADM Preliminary Phase, and how it relates to the Content Framework and the Architecture Development Method (ADM). The chapter spans printed pages 63-110.

Detailed Section Breakdown

5.1 Sections of the Enterprise Metamodel (pp. 65-66)

The Enterprise Metamodel is described as a **Foundation Enterprise Metamodel** (p. 63) — since no two organizations are alike, it must be tailored to organizational needs; the version in the TOGAF Standard is a starting point, not a final product. The Content Framework provides a categorization mechanism to structure a representation of the Enterprise Metamodel, consisting of **five interrelated sections**, and this must not be confused with an architecture diagram showing domains or layers (p. 63).

The Enterprise Metamodel comprises five interconnected **sections** (the word "sections" is used deliberately because the Enterprise Metamodel is a metamodel, not an architecture diagram) (p. 64):

- General Entities
- Business Architecture

- Data Architecture
- Application Architecture
- Technology Architecture

These correspond to the four architecture domains (Business, Data, Application, Technology) plus one generic section. Figure 5-1 (p. 65), "The Foundation Enterprise Metamodel," shows all sections together; relations between entities were intentionally omitted from Figures 5-1, 5-2, 5-5, 5-14, 5-17, and 5-19 to improve readability (p. 64). Each section consists of a set of model entities — architecture concepts each with its own definition and explanation; understanding them is critical to understanding the architecture products described later in the book (p. 65). All examples throughout the chapter refer to a fictional healthcare organization (p. 67), used consistently in subsequent chapters as well.

5.1.1 General Entities (pp. 65-73)

The upper section of the Enterprise Metamodel, called **General Entities**, contains **eight entities** (p. 65), shown in Figure 5-2 (p. 66) and Table 5-1 (pp. 66-67). These entities are so generic that they cannot be categorized into any one of the four core architecture domains — they can be used/related across all four domains (e.g., *principle* can relate to a Business Architecture or Technology Architecture element; *location* can relate to an application component or a role/actor) (p. 67).

The eight General Entities and their table definitions (Table 5-1, pp. 66-67):

- **Principle** — a qualitative statement of intent that should be met by the architecture, with at least a descriptive name, supporting rationale, and one or more implications.
- **Constraint** — an external factor that prevents an organization from pursuing particular approaches to meet its goals.
- **Requirement** — a quantitative statement of business need that must be met by a particular architecture or work package.
- **Assumption** — a statement of probable fact that has not been fully validated at this stage, due to external constraints.
- **Location** — a place where activities occur; locations can be composed and decomposed.
- **Gap** — a statement of difference between two states, used in gap analysis (Baseline vs. Target Architecture).

- **Work Package** — a set of actions identified to achieve one or more business objectives; can be part of a project, a complete project, or a program.
- **Capability** — an ability that an organization, person, or system possesses.

Expanded explanations follow (pp. 68-73):

Principle (p. 68): Described as similar to the constitution of a country — highly abstract, strategic, carefully designed to fit the organization/business model, and (like a constitution) changes little and has a strong binding character. There are **Architecture Principles** and **Enterprise Principles** (cross-referenced to Chapter 14, Section 14.2). Basic/high-level principles derive from business goals and objectives.

Constraint (p. 68): Mostly external; example given — wanting to create a new business role/function but being blocked because an external staffing party enforces standard job profiles that don't accommodate it.

Requirement (p. 68): Indicates changes needed to a new/existing situation, process, or system within an initiative (work package); represents essential attributes/behaviors a system/architecture/solution must have; guides design, development, and validation.

Figure 5-3 (p. 69), "Relationship between a principle, requirement, and constraint," shows two principles feeding into one Objective ("Make adjustments to comply with legislation demands by Q3 2028"): the left principle ("Use standard products and services") relates to a Constraint ("Only existing job profiles can be used") which relates to an Initiative ("Fit job profile X into existing job profile"); the right principle ("Comply with legislation") relates to a Requirement ("Job profile X is needed because of new legislation") which relates to another Initiative ("Recruit staff for job profile X"). This demonstrates that requirements *detail* principles (enabling them) while constraints *limit* principles (though an initiative may try to remove the constraint).

Assumption (pp. 69-70): A foundational element defining/communicating uncertainties or information gaps that could influence architecture decisions; necessary when decisions must proceed despite incomplete knowledge (e.g., assuming an existing application will support certain functional requirements before those are individually validated). The TOGAF Series Guide "Architecture Project Management" elaborates on formulating assumptions.

Location (p. 70): A place where activity happens, **literally** — physical locations (office buildings, countries, continents); also used for information systems/applications/technology components, e.g., diagrams showing distribution of server systems across data centers.

Gap (p. 70): Difference/shortfall between the **Baseline Architecture** (current state) and **Target Architecture** (desired future state — a footnote clarifies this "future state" is not definitive, just a state at a specific point in time). Details the results of gap analysis performed in **Phases B, C, and D** of the ADM; an important input for implementation/migration planning.

Figure 5-4 (p. 71), "Gaps between Baseline, Transition, and Target Architectures": illustrates replacing Application A with Application B via a phased approach — Baseline Architecture ("Application A needs to be replaced by Application B") → Gap ("Application A misses necessary functionality") → Transition Architecture ("Application B is implemented alongside Application A") → Gap ("Processes are pointing towards Application A") → Target Architecture ("Application A is replaced by Application B").

Work Package (pp. 71-72): Also called an **initiative** — a planned action/project to achieve specific goals; represents actionable steps (projects, programs, activities) implementing architectural change; also used to represent activities required to close gaps moving from Baseline through Transition to Target Architecture.

Capability (pp. 72-73): Central to **capability-based planning**, an EA approach focusing on what an organization is able to **do** rather than what it **has**. Examples: an airplane's capability to fly (unique vs. a car's shared capability to transport); a motorcycle's capability to maneuver between cars (requires the rider to actively perform it, unlike an airplane's autopilot). Capabilities define what an organization/person/object **should** be able to do; actually executing them requires people, processes, applications, and/or technology. Important distinction (p. 73): the metamodel also has a separate **Business Capability** entity (defined in 5.1.2) — a business capability differs from a plain capability by having an **actionable component**: "to achieve a specific purpose or outcome" [13]. A plain capability lacks this purpose qualifier, making business capabilities more meaningful/valuable than generic capabilities.

Further Reading (p. 73): TOGAF Series Guide "Architecture Project Management" (document number **G188**).

5.1.2 Business Architecture (pp. 74-92)

The most comprehensive section of the Enterprise Metamodel — **19 entities in total** (p. 74), shown in Figure 5-5 (p. 74) and Table 5-2 (pp. 74-76).

Table 5-2 entities and definitions:

- **Driver** — external/internal condition motivating the organization to define its goals.
- **Goal** — high-level statement of intent/direction, typically used to measure organizational success.
- **Objective** — organizational aim declared in a SMART (Specific, Measurable, Achievable, Realistic, Timebound) way.
- **Measure** — an indicator/factor tracked (usually ongoing) to determine success/alignment with objectives and goals.
- **Course of Action** — direction/focus provided by strategic goals/objectives, often to deliver the value proposition in the business model.
- **Business Capability** — a particular ability a business may possess/exchange to achieve a particular purpose.
- **Value Stream** — representation of an end-to-end collection of activities creating an overall result for a customer/stakeholder/end user.
- **Function** — a set of business behaviors based on chosen criteria; usually close-coupled to organizational units.
- **Process** — a sequence of activities achieving a specified outcome; can be decomposed into sub-processes; shows operation of a business capability or service.
- **Business Service** — supports the business by encapsulating a unique element of business behavior; external services may be supported by business services.
- **Organization Unit** — a self-contained unit of resources with goals, objectives, and measures; may include external parties/business partners.
- **Actor** — a person, organization, or system with a role that initiates/interacts with activities; internal or external.
- **Role** — the usual/expected behavior of an actor, or the part played in a process/event; an actor may have multiple roles.
- **Product** — an outcome generated by the business offered to customers; includes materials and/or services.
- **Contract** — an agreement between consumer and provider establishing functional and non-functional interaction parameters; applies to all service interactions in the metamodel.

- **Business Information** — represents a concept and its semantics used within the business.
- **Control** — a decision-making step with decision logic determining process execution approach or governance compliance.
- **Event** — an organizational state change that triggers processing events; may originate/resolve inside or outside the organization.
- **Service Quality** — a configuration of non-functional requirements/attributes assignable to a business, application, or technology service.

Expanded explanations (pp. 77-92):

Driver (p. 77): Any influence/factor/condition pushing the organization toward decisions/actions; often the *reason* behind change. May arise externally (market share, legislation, technological advancements) or internally (leadership changes, operational inefficiencies). When stakeholder-related, drivers are called **concerns**.

Goal (p. 77): Can be anything a stakeholder wants — a state of affairs or value produced. Examples: increase profits, reduce help desk wait times, implement new regulatory requirements. Usually expressed qualitatively (increase, improve, easier).

Objective (p. 77): Further details a goal in SMART terms; example — goal "implement new regulatory requirements" relates to objective "make adjustments to comply with legislation demands by Q3 2028."

Measure (p. 78): A quantitative indicator tracking progress toward a goal/objective; provides evidence-based insight into performance. **Three types of measures:**

- **Key Performance Indicators (KPIs)**: specific measures tied to business outcomes and Critical Success Factors (CSFs); outcome-focused, monitor progress against strategic goals.
- **Metrics**: broader measures (quantitative and qualitative); not always tied directly to outcomes; measure process efficiency, resource usage, operational aspects.
- **Target Values**: associated desired performance levels providing a reference point for evaluating current performance.

Course of Action (pp. 79-80): Represents what an organization has *decided to do* — a coordinated set of activities, decisions, and policies. Divided into **strategies** (long-term, broad

scope) and **tactics** (shorter-term, narrower scope) — no sharp line between the two. Figure 5-6 (p. 79), "Relationship between a driver, goal, objective, measure, and course of action": Driver "Legislation" → Goal "Implement new regulatory requirements" ← Objective "Make adjustments to comply with legislation demands by Q3 2028"; Measure "Q3 Legal Department Report" feeds the Goal; Course of Action "Acquire employees" relates to the Goal and loops back to the Objective.

Business Capability (p. 80): Cross-referenced to the Capability discussion in 5.1.1 (p. 73).

Value Stream (pp. 80-83): Represents flow of activities delivering value to internal/external stakeholders; not necessarily linear. Each value stream consists of **stages** — high-level phases of value-delivering activity. Value streams are *supported by* business capabilities. Key distinction: **value streams vs. business processes** — a business process describes the (time-ordered) sequence of behaviors for an individual case, including alternative paths/decision points; a value stream focuses on overall value-creating behavior from the perspective of importance/worth/usefulness of the output, and is *not* a description of time-ordered tasks for individual cases. Value streams reflect an organization's **business model**; business processes reflect its **operating model** (p. 80-81). Key principle: value is always defined **from the perspective of the stakeholder** — "the value obtained is in the eye of the beholder" [14] (p. 82). Value streams never deliver a tangible end product — they deliver *value* (contrast: a business process's end product, e.g., a signed employee agreement, IS tangible) (p. 82).

Example — value stream "Acquire Talent" (Figure 5-7, p. 81) has three stages:

1. **Sourcing and Attraction** — identify/attract candidates; process steps: creating/posting job descriptions, reaching out to candidates, building a talent pipeline (includes employer branding).
2. **Selection and Assessment** — evaluate/select suitable candidates; process steps: screening resumes, conducting interviews/assessments, narrowing the candidate pool.
3. **Offer and Onboarding** — finalize hiring, integrate new hire; process steps: extending job offers, negotiating terms, onboarding the new hire.

Figure 5-8 (p. 82) shows these stages populated with the corresponding process steps. The value produced is "hired talent" (intangible).

Function (p. 83): A fundamental building block — a distinct, cohesive set of activities/services contributing to an outcome. Composed of processes, procedures, and interactions. Put simply: a

logical *grouping of activities* supporting overall business operation. Functions often reflect organizational structure but are tied to activities, not directly to departments.

Process (pp. 83-84): A set of related, sequential activities/tasks achieving a specific goal/output. Figure 5-9 (p. 84) shows three parent processes matching the value stream stages: **Acquiring Talent, Evaluating Candidates, Employee Hiring**, each containing its series of activities. **Function vs. Process distinction** (p. 84, Figure 5-10): function is broader/more abstract than process — "processes define *how* things get done step by step, while functions define *what* activities are necessary" to support organizational goals. Figure 5-10 example: Function "Talent Acquisition" contains Process "Acquiring Talent"; Function "Recruitment" contains Process "Evaluating Candidates"; Function "Human Resources" contains Process "Employee Hiring."

Organization Unit (pp. 84-85): Comparable to a department; similar to a business actor per TOGAF logic. Important nuance: when considering goals/objectives/measures, the element closest to the framework's intent for organization unit is the **stakeholder** — a stakeholder is "a role of an individual, team, or organization (or classes thereof) that represents their interests in the outcome of the architecture." So an organization unit is best represented *as a stakeholder* for relationships to goals/objectives/measures.

Actor (p. 85): An entity interacting with or affected by an Enterprise Architecture (direct or indirect); typically external to the architecture itself. Key distinction from **Organization Unit**: an actor CAN represent an entire organization but NOT a department within it — representing a department is exclusive to organization unit. Actors usually represent individuals within an organization or external parties.

Role (pp. 85-86): Describes responsibilities, behaviors, expectations tied to a position/function; linked to tasks/duties performed; can span business, data, application, and technology domains. Figure 5-11 (p. 86), "Actors within an organization unit with an assigned role": Organization Unit "Human Resources (HR)" contains two Actors (John H. Smith and Nina Hunter), both sharing the Role "HR Business Partner."

Product (p. 86): An outcome of a process, always of value to a customer; includes materials and/or services, often accompanied by a contract. Example: employment agreement is the product of the hiring process, offered to the employee (the customer); terms of employment set out in the accompanying contract. Can be compared to a **business object** (a concept used within a business domain). Figure 5-12 (p. 86), "A process creating an end product accompanied by a contract":

Process "Employee Hiring" with steps "Extending job offer" → "Negotiating terms" → "Onboarding the new hire," generating business objects "Job Offer" and "Terms," culminating in Product "Employment Agreement" plus Contract "Terms of Employment."

Contract (p. 87): A formal agreement outlining expectations, responsibilities, obligations between parties; formalizes agreements related to architecture development/implementation; applies to all service interactions in the metamodel.

Business Information (pp. 87-89): Concerned with data/information necessary for business operations and decision-making. Links to the **information concept** from the Business Architecture Body of Knowledge (**BIZBOK**) Guide, defined as: "An information concept is comprehensive in terms of representing objects across a business ecosystem, but may or may not be defined in a data model or database. The information concept realizes or makes explicit a business object, with the object serving as the basis for the information concept name" [5] (p. 88). Information concepts are high-level abstractions (not raw data) representing meaningful business information (e.g., "strategy and vision," "human resource and competency"), independent of specific systems/technology. Notably: **the Information Architecture domain disappeared from the TOGAF Standard after version 8 (2002)**, but the author notes a possible movement toward its return (p. 87). Per TOGAF, a **data entity** (Section 5.1.3) corresponds to the information concept from BIZBOK (p. 89).

Control (pp. 89-90): Mechanisms/processes ensuring architecture aligns with strategic goals and operates effectively. Used for five situations:

- **Governance** — establishing structures/processes to guide/oversee EA development and implementation (roles, responsibilities, decision-making).
- **Compliance** — ensuring architecture adheres to regulations/standards/policies via checks/audits.
- **Monitoring and Reporting** — tracking performance/progress via metrics, monitoring systems, reporting structures.
- **Risk Management** — identifying, assessing, mitigating architecture-related risks.
- **Change Management** — managing architecture changes in a controlled, systematic way (impact assessment, coordination, alignment with vision/strategy).

Event (pp. 90-91): An organizational state change with notable impact/significance (change in market conditions, new regulation, hiring). Can be triggered externally (new laws, competitive actions) or internally (process changes, project milestones). Figure 5-13 (p. 90), "Event triggering a

process": Event "Candidate Selected" triggers Process "Employee Hiring" (steps: Extending job offer → Negotiating terms → Onboarding the new hire).

Service Quality (p. 91): Measures how well a service meets/exceeds stakeholder expectations (reliability, responsiveness, user satisfaction). Tied to **IT service management**; emphasized particularly in the Technology Architecture domain regarding Solution Architecture.

Business Service (pp. 91-92): A logical, self-contained unit of business functionality valuable to the organization/customers; may serve external or internal stakeholders, who don't need to know internal delivery details. **Critical structural role:** the business service is the **link/layer between Business Architecture and the Data, Application, and Technology Architectures** — the interface between operational architectures and strategic/tactical architectures.

Further Reading (p. 92): "Business Streams And Value Processes" blog (eawheel.com); a YouTube video on Business Architecture and value streams; the BIZBOK Guide (businessarchitectureguild.org); TOGAF Series Guide "Information Mapping," document number **G190**.

5.1.3 Data Architecture (pp. 92-95)

Described as directly related to the **business service** entity (the connecting concept). Contains **three entities** (Figure 5-14, p. 92; Table 5-3, p. 93):

- **Data Entity** — data recognized by the business as a distinct concept.
- **Logical Data Component** — a data structure composed of logically related data entities.
- **Physical Data Component** — a data structure realizing related logical data components in the format/schema required by a particular technology.

Data Entity (pp. 93-94): Typically represented via entity-relationship diagrams (ERDs) or class diagrams, showing relationships, constraints, attributes. Closely tied to Business Architecture; helps define **Business Information Models**. Has **three relationships** with Business Architecture entities: (1) accessed/updated by a business service; (2) consumed by an actor; (3) **realizes** a business information entity. This realization relationship transforms data into valuable information — illustrated by the example of raw numbers (22, 23, 22, 25, 21) gaining value/meaning only once

labeled as "expected temperature for the next five days" (Figure 5-15, p. 94: Data Entity → Business Information; "Data + Context = Valuable information").

Logical Data Component (pp. 94-95): Abstract, technology-agnostic structure representing what data is and how it relates to business operations, independent of storage/implementation.

Physical Data Component (p. 95): Tangible implementation of data systems — hardware/software infrastructure (databases, file servers, data warehouses, cloud storage) realizing logical data components. Figure 5-16 (p. 95) chain example: Physical Data Component "Database System" → Logical Data Components "Candidate Name" and "Candidate Experience" → Data Entity "Candidate Information" → Business Information "Candidate Profile."

5.1.4 Application Architecture (pp. 96-98)

Second of the three architectures at the bottom of the Content Framework; contains **three entities** (Figure 5-17, p. 96; Table 5-4, p. 96):

- **Application Service** — automated elements of a business service; may deliver/support part or all of one or more business services.
- **Logical Application Component** — an encapsulation of application functionality definable by services offered and data maintained, independent of implementation/technology.
- **Physical Application Component** — a realization of logical application functionality using components that may be hired, procured, or built.

Application Service (p. 97): Delivers specific functionality an organization needs, supporting business processes via automated/semi-automated functions.

Logical Application Component (pp. 97-98): Abstract representation of software performing specific functions/services, encapsulating behavior/responsibilities without physical implementation details — focuses on *what* the system should do, not *how*. Example (Figure 5-18, p. 97): a logical component may represent an HR recruitment system handling "storing candidate information and tracking interviews" without specifying whether it's Recruitment Application A, B, or custom-built (that decision belongs to the physical design phase). Chain: Physical Application Component "HR Recruitment Application" → Logical Application Component "Storing Candidate Information" → Application Service "Data Storing Service" → Business Service

"Candidate Evaluation Service," with Data Entity "Candidate Information" linked to the application service.

Physical Application Component (p. 98): Actual, concrete implementation of an application/software running on a specific technology platform — tangible aspects including installed software, configurations, and underlying hardware.

5.1.5 Technology Architecture (pp. 98-101)

Third of the three architectures at the bottom of the Content Framework; contains **three entities** (Figure 5-19, p. 98; Table 5-5, p. 99):

- **Technology Service** — a technical capability required to provide enabling infrastructure supporting application delivery.
- **Logical Technology Component** — an implementation-independent encapsulation of technology services.
- **Physical Technology Component** — a realization of logical technology functionality using a particular deployable technology product.

Technology Service (p. 99): Provides technical capabilities from underlying infrastructure supporting higher-level business/application/data services. Example: a cloud storage service providing scalable, secure, on-demand storage (file storage, data backups, accessibility from multiple locations).

Logical Technology Component (pp. 99-100): Abstraction representing a technology-related function/capability without specific physical implementation details — focuses on functionality required, not the actual hardware/software/network. Example: a **Database Management System (DBMS)** logical component doesn't specify MySQL, Oracle, or SQL Server — just indicates the need for structured data management. In the running example, the fictitious company's DBMS manages structured candidate data, supports SQL queries, ensures data integrity, and provides backups, without naming a vendor.

Physical Technology Component (pp. 100-101): Actual tangible IT system parts (hardware, software, networking devices) deployed in an organization's Technology Architecture. Figure 5-20 (p. 100) chain: Physical Technology Component "Database System" → Logical Technology

Component "Database Management System" → Technology Service "Cloud Storage Service" → Physical Application Component "HR Recruitment Application" (illustrating how, if MySQL is selected, it realizes the logical DBMS component).

Further Reading (p. 101): "TOGAF Content Framework and Enterprise Metamodel" chapter of the TOGAF Standard; "Business Capabilities, Version 2" Series Guide, document number **G211**.

5.2 Tailoring the Enterprise Metamodel (pp. 101-107)

The Enterprise Metamodel, the **Content Framework**, and the **Architecture Development Method (ADM)** are described as "inextricably linked" — the ADM is often considered the heart of the TOGAF Standard and lets architects create an **organization-specific version** of the Enterprise Metamodel. This tailoring occurs during the **Preliminary Phase** (cross-referenced to Chapter 12, Section 12.3) (p. 101).

Figure 5-21 (p. 102), "Relation between ADM, Enterprise Metamodel, and Content Framework": shows the Preliminary Phase of the ADM developing the organization-specific metamodel; the Enterprise Metamodel supplies entities to the Content Framework; the Content Framework in turn further develops/evolves the Metamodel and complements the ADM; the ADM updates the Content Framework. A note clarifies the arrows represent relationships **between models**, not connections to specific ADM phases (p. 102).

Key points on the tailoring process (pp. 102-106):

- The Enterprise Metamodel, once formed (organization-specific), serves as **input to the Content Framework**, which describes and structures architecture using the metamodel's entities to create deliverables/artifacts, and which in turn enables the Metamodel's evolution (p. 103).
- The TOGAF Standard explicitly does **not** aim to constrain an enterprise's selection of artifacts [2] (p. 103).
- Organization-specific metamodel development happens during the **Preliminary Phase**, when the **Enterprise Architecture Capability** is being established; architects may exclude irrelevant entities/relationships and/or add new ones [2] (p. 103).
- The author's own recommendation/critique (pp. 104-105): although not explicit in TOGAF, the **organization unit** entity is effectively used to refer to **stakeholders** (consistent with TOGAF's definition: "a self-contained unit of resources with goals, objectives, and measures"). The author strongly suggests explicitly **adding a "stakeholder" entity** to the Enterprise Metamodel,

noting that **ArchiMate** *does* recognize a Stakeholder element and recommends using it when TOGAF's Organization Unit relates to Measure, Goal, or Objective [15] — this integrates TOGAF with ArchiMate (both Open Group standards).

- Other entities recommended for addition to an organization-specific metamodel (p. 105): **stakeholder, business collaboration, business interaction, business interface, resource, risk, and value**. The stakeholder entity is used in the Stakeholder Catalog; most others appear in the Business Model Diagram (cross-referenced to Chapter 12, Section 12.4.2).
- The first ADM iteration focuses on creating the Architecture Capability, which includes developing an **initial version of the Enterprise Metamodel** — an organization-specific variant. The author emphasizes this step is frequently overlooked by architects (p. 104).

Five-step process for tailoring the Enterprise Metamodel (pp. 106, restated pp. 109-110):

4. **Understand the Enterprise Metamodel** — get familiar with the Foundation Enterprise Metamodel's standard entities/relationships.
5. **Customize the Metamodel** — identify and include exceptional entities unique to the organization.
6. **Integrate the Customized Metamodel** — use it as input to the Content Framework, which defines all relevant architecture deliverables (artifacts, building blocks); add/remove deliverables to customize the framework.
7. **Align Deliverables** — create architecture deliverables per the customized Content Framework as you progress through ADM phases.
8. **Continuously Evolve** — regularly review/update both the Enterprise Metamodel and Content Framework as the organization and its environment change.

The author stresses (p. 106): "Tailoring the Enterprise Metamodel is an absolute necessity. The foundation model provided in the TOGAF standard is only a starting point."

Further Reading (p. 107): "The TOGAF® Leader's Guide to Establishing and Evolving an EA Capability," Chapter 8 ("Customization of Architecture Contents and Metamodel"), document number **G184**; "How to Use the ArchiMate® Modeling Language to Support the TOGAF® Standard," document number **G21E**.

5.3 Summary (pp. 107-110)

Recaps that the Enterprise Metamodel is a core TOGAF component providing context for artifacts referenced across ADM phases, and that it and the Content Framework are kept in sync (metamodel supplies entities; framework evolves the metamodel). Restates the five sections (General Entities, Business Architecture, Data Architecture, Application Architecture, Technology Architecture) (p. 107).

Key numeric facts reiterated (p. 108):

- **General Entities: 8 entities** total.
- **Business Architecture: 19 entities** total — with a notable framework comparison: TOGAF places driver, goal, objective, and measure all under one heading, **Business Architecture**, whereas **ArchiMate** assigns these same concepts to **motivation elements**. Reason given: TOGAF has no distinct motivational/strategic element category — it organizes everything into the four architecture domains (Business, Data, Application, Technology).
- **Data Architecture, Application Architecture, and Technology Architecture: 3 entities each.**
- **Total across the whole Enterprise Metamodel: 36 entities** ($8 + 19 + 3 + 3 + 3 = 36$), all described with practical examples in the chapter.

Restates that the Foundation Enterprise Metamodel is a starting point, entities can be added/removed per organizational needs, and that the Metamodel, Content Framework, and ADM are inextricably linked (organization-specific metamodel creation occurs in the Preliminary Phase). Restates the five-step tailoring process (Understand, Customize, Integrate, Align, Continuously Evolve) in full (pp. 109-110), concluding: "The Enterprise Metamodel is the foundation of the Enterprise Architecture. It takes time and care to create a customized and organization-specific version of the metamodel. The model serves as input to both the Content Framework and the Architecture Development Method" (p. 110).

Key Terms & Definitions

General Entities (8) — p. 65

- **Principle** (p. 66) — A qualitative statement of intent that should be met by the architecture; has at least a descriptive name, supporting rationale, and one or more implications. (Expanded, p. 68: "at least a supporting rationale and a measure of importance.")
- **Constraint** (p. 66) — An external factor that prevents an organization from pursuing particular approaches to meet its goals.

- **Requirement** (p. 66) — A quantitative statement of business need that must be met by a particular architecture or work package.
- **Assumption** (p. 66) — A statement of probable fact that has not been fully validated at this stage, due to external constraints.
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- **Gap** (p. 67) — A statement of difference between two states; used in gap analysis between Baseline and Target Architecture.
- **Work Package** (p. 67) — A set of actions identified to achieve one or more objectives for the business; can be part of a project, a complete project, or a program. (Also called an **initiative**, p. 71.)
- **Capability** (p. 67) — An ability that an organization, person, or system possesses.

Business Architecture Entities (19) — p. 74

- **Driver** (p. 74) — An external or internal condition that motivates the organization to define its goals.
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- **Organization Unit** (p. 75) — A self-contained unit of resources with goals, objectives, and measures; may include external parties and business partner organizations.
- **Actor** (p. 76) — A person, organization, or system that has a role that initiates or interacts with activities; internal or external.

- **Role** (p. 76) — The usual or expected behavior of an actor, or the part played in a particular process or event; an actor may have a number of roles.
- **Product** (p. 76) — An outcome generated by the business to be offered to customers; includes materials and/or services.
- **Contract** (p. 76) — An agreement between a consumer and a provider that establishes functional and non-functional parameters for interaction; applies to all service interactions in the metamodel.
- **Business Information** (p. 76) — Represents a concept and its semantics used within the business.
- **Control** (p. 76) — A decision-making step with accompanying decision logic, used to determine execution approach for a process or ensure governance compliance.
- **Event** (p. 76) — An organizational state change that triggers processing events; may originate/be resolved inside or outside the organization.
- **Service Quality** (p. 76) — A configuration of non-functional requirements or attributes assignable to a business, application, or technology service.

Data Architecture Entities (3) — p. 93

- **Data Entity** (p. 93) — Represents data that is recognized by the business as a distinct concept.
- **Logical Data Component** (p. 93) — A data structure composed of logically related data entities.
- **Physical Data Component** (p. 93) — A data structure that realizes related logical data components in the format/schema required by a particular technology.

Application Architecture Entities (3) — p. 96

- **Application Service** (p. 96) — The automated elements of a business service; may deliver/support part or all of one or more business services.
- **Logical Application Component** (p. 96) — An encapsulation of application functionality definable by services offered and data maintained, independent of implementation and technology.
- **Physical Application Component** (p. 96) — A realization of logical application functionality using components that may be hired, procured, or built.

Technology Architecture Entities (3) — p. 99

- **Technology Service** (p. 99) — A technical capability required to provide enabling infrastructure that supports the delivery of applications.

- **Logical Technology Component** (p. 99) — An implementation-independent encapsulation of technology services.
- **Physical Technology Component** (p. 99) — A realization of logical technology functionality using a particular technology product that may be deployed.

Other Key Terms

- **Foundation Enterprise Metamodel** (p. 63/65) — The standard, un-tailored version of the Enterprise Metamodel as presented in the TOGAF Standard; meant as a starting point to be customized.
- **Concerns** (p. 77) — Term used when drivers are stakeholder-related.
- **KPI (Key Performance Indicator)** (p. 78) — A specific measure tied to business outcomes and Critical Success Factors (CSFs); outcome-focused.
- **Metrics** (p. 78) — Broader measures (quantitative/qualitative) not always tied directly to outcomes; measure process efficiency, resource usage.
- **Target Values** (p. 78) — Associated desired performance levels for a measure, serving as a reference point.
- **Stages** (of a value stream) (p. 80) — High-level phases/categories of activities performed to deliver value.
- **Business Model** vs. **Operating Model** (pp. 80-81) — Value streams reflect an organization's business model and value proposition; business processes reflect its operating model.
- **Stakeholder** (p. 85, 104) — A role of an individual, team, or organization (or classes thereof) that represents their interests in the outcome of the architecture; author recommends adding this explicitly as an entity (TOGAF does not formally include it, but ArchiMate does).
- **Business Object** (p. 86) — A concept used within a particular business domain; a product can sometimes be compared to a business object.
- **Information Concept** (BIZBOK) (p. 88) — A high-level abstraction representing a fundamental piece of information a business uses/manages/interacts with; corresponds to TOGAF's "data entity."
- **Baseline Architecture** (p. 70) — The current state of the Enterprise Architecture.
- **Target Architecture** (p. 70) — The desired future state of the Enterprise Architecture (at a specific point in time, not definitive).
- **Transition Architecture** (p. 71) — An intermediate architecture state between Baseline and Target.
- **Capability-Based Planning** (p. 72) — An EA approach focusing on what an organization is able to do, rather than what it has.
- **BIZBOK Guide** (p. 87) — Business Architecture Body of Knowledge Guide; source of the "information concept" definition linked to TOGAF's business information entity.

Figures & Tables Referenced

- **Figure 5-1** (p. 65) — "The Foundation Enterprise Metamodel." Shows all five sections together (General Entities, Business/Data/Application/Technology Architecture) in one diagram; the master overview figure for the whole chapter.
- **Figure 5-2** (p. 66) — "General Entities of the Enterprise Metamodel." Shows the eight General Entities and notes they are "Associated with all objects."
- **Table 5-1** (pp. 66-67) — "General Entities." Lists all 8 General Entities with descriptions.
- **Figure 5-3** (p. 69) — "Relationship between a principle, requirement, and constraint." Shows two principles feeding an objective, one linked to a constraint/initiative, the other to a requirement/initiative — illustrates how principles are detailed by requirements and limited by constraints.
- **Figure 5-4** (p. 71) — "Gaps between Baseline, Transition, and Target Architectures." Illustrates a phased application replacement (App A to App B) using the gap entity at each transition point.
- **Figure 5-5** (p. 74) — "Business Architecture Entities of the Enterprise Metamodel." Shows all 19 Business Architecture entities in their metamodel layout.
- **Table 5-2** (pp. 74-76) — "Business Architecture Entities." Lists all 19 entities with descriptions.
- **Figure 5-6** (p. 79) — "Relationship between a driver, goal, objective, measure, and course of action." Ties together the strategic/motivational chain using the legislation-compliance example.
- **Figure 5-7** (p. 81) — "Value stream stages of Acquire Talent." Shows the three stages (Sourcing and Attraction, Selection and Assessment, Offer and Onboarding) from Stakeholder to Hired Talent.
- **Figure 5-8** (p. 82) — "Value stream supplemented by business processes." Adds the underlying process steps beneath each value stream stage.
- **Figure 5-9** (p. 84) — "Processes containing series of activities." Shows three parent processes (Acquiring Talent, Evaluating Candidates, Employee Hiring), each containing its activity steps.
- **Figure 5-10** (p. 84) — "Functions containing corresponding processes." Illustrates function-process nesting (Talent Acquisition/Recruitment/Human Resources functions containing their respective processes) — key for the function vs. process distinction.
- **Figure 5-11** (p. 86) — "Actors within an organization unit with an assigned role." Shows HR organization unit containing two actors sharing one role (HR Business Partner).
- **Figure 5-12** (p. 86) — "A process creating an end product accompanied by a contract." Shows the Employee Hiring process generating business objects, a product (Employment Agreement), and a contract (Terms of Employment).

- **Figure 5-13** (p. 90) — "Event triggering a process." Shows the event "Candidate Selected" triggering the Employee Hiring process.
- **Figure 5-14** (p. 92) — "Data Architecture Entities of the Enterprise Metamodel." Shows the three Data Architecture entities.
- **Table 5-3** (p. 93) — "Data Architecture Entities." Lists Data Entity, Logical Data Component, Physical Data Component with descriptions.
- **Figure 5-15** (p. 94) — "Relationship between the data entity and business information." Illustrates "Data + Context = Valuable information" using the temperature-numbers example.
- **Figure 5-16** (p. 95) — "Physical and logical data components in relation to a data entity and business information." Full chain example: Database System → Candidate Name/Experience (logical) → Candidate Information (data entity) → Candidate Profile (business information).
- **Figure 5-17** (p. 96) — "Application Architecture Entities of the Enterprise Metamodel." Shows the three Application Architecture entities.
- **Table 5-4** (p. 96) — "Application Architecture Entities." Lists Application Service, Logical Application Component, Physical Application Component with descriptions.
- **Figure 5-18** (p. 97) — "Relationship between physical and logical application components and application and business services." Full chain: HR Recruitment Application → Storing Candidate Information → Data Storing Service → Candidate Evaluation Service (business service), linked to the Candidate Information data entity.
- **Figure 5-19** (p. 98) — "Technology Architecture Entities of the Enterprise Metamodel." Shows the three Technology Architecture entities.
- **Table 5-5** (p. 99) — "Technology Architecture Entities." Lists Technology Service, Logical Technology Component, Physical Technology Component with descriptions.
- **Figure 5-20** (p. 100) — "Relationship between physical and logical technology components and technology and application services." Full chain: Database System (physical) → Database Management System (logical) → Cloud Storage Service (technology service) → HR Recruitment Application (physical application component).
- **Figure 5-21** (p. 102) — "Relation between ADM, Enterprise Metamodel, and Content Framework." Central figure for Section 5.2, showing how the Preliminary Phase of the ADM develops the organization-specific metamodel, and the bidirectional relationships among ADM, Enterprise Metamodel, and Content Framework.

Key Takeaways / Exam Focus Points

- The Enterprise Metamodel has **five interrelated sections**: General Entities, Business Architecture, Data Architecture, Application Architecture, Technology Architecture (p. 64).

- Total entity counts: **General Entities = 8; Business Architecture = 19; Data Architecture = 3; Application Architecture = 3; Technology Architecture = 3; grand total = 36 entities** (pp. 65, 74, 93, 96, 99, 108).
- The Enterprise Metamodel, Content Framework, and ADM are described as an inseparable "triangle" — metamodel supplies entities to the Content Framework; Content Framework evolves the metamodel; the ADM tailors the metamodel during the **Preliminary Phase** (p. 101).
- **Capability** (General Entity, p. 67/72) vs. **Business Capability** (Business Architecture entity, p. 75/80): the key difference is that business capability includes an **actionable component** — "to achieve a specific purpose or outcome" — making it more meaningful than a plain capability (p. 73).
- **Three types of Measures:** KPIs, Metrics, Target Values (p. 78) — a classic enumeration likely to be tested.
- **Value Stream vs. Business Process:** value streams = business model/value proposition perspective, non-linear, deliver intangible value; business processes = operating model perspective, time-ordered, individual-case-oriented, may produce tangible outputs (pp. 80-82).
- **Function vs. Process:** function is broader/more abstract ("what" activities are necessary); process is the specific step-by-step sequence ("how" things get done) (p. 84).
- **Actor vs. Organization Unit:** an actor CAN represent an entire organization but NOT a department; representing a department is reserved for organization unit (p. 85).
- **Organization Unit** is best treated as a **stakeholder** when related to goals, objectives, and measures — even though TOGAF doesn't use the word "stakeholder" formally in the base metamodel (p. 85, 104).
- **Business Service** is the critical link/interface layer connecting Business Architecture to Data, Application, and Technology Architectures (p. 91).
- Data/Application/Technology Architecture sections each follow the same three-tier pattern: **[Domain] Service → Logical [Domain] Component → Physical [Domain] Component**, moving from business-facing abstraction to technology-agnostic logical design to concrete/tangible implementation (pp. 93, 96, 99).
- **Gap analysis** occurs specifically in **ADM Phases B, C, and D** (p. 70).
- Author's key critique/addition: TOGAF's base metamodel lacks an explicit **stakeholder** entity; the author recommends adding stakeholder, business collaboration, business interaction, business interface, resource, risk, and value entities when tailoring (p. 105). Stakeholder is used in the Stakeholder Catalog; most others appear in the Business Model Diagram.
- **ArchiMate comparison:** TOGAF groups driver, goal, objective, measure under "Business Architecture," while ArchiMate classifies these as **motivation elements** — because TOGAF has no separate motivational/strategic category, only the four architecture domains (p. 108).
- **Five-step tailoring process:** (1) Understand the Enterprise Metamodel, (2) Customize the Metamodel, (3) Integrate the Customized Metamodel, (4) Align Deliverables, (5) Continuously Evolve (pp. 106, 109-110).

- The **Information Architecture domain** disappeared from TOGAF after version 8 (2002); the author speculates it may be returning, given the tie between business information and BIZBOK's information concept (p. 87).
- SMART = **S**pecific, **M**easurable, **A**chievable, **R**ealistic, **T**imebound (p. 77, footnote 2).
- The TOGAF Standard explicitly does **not** aim to constrain an enterprise's selection of artifacts (p. 103).

Self-Check Questions

Q: What are the five interrelated sections of the Enterprise Metamodel?

A: General Entities, Business Architecture, Data Architecture, Application Architecture, and Technology Architecture. (p. 64)

Q: How many entities are in each section of the Enterprise Metamodel, and what is the total?

A: General Entities = 8, Business Architecture = 19, Data Architecture = 3, Application Architecture = 3, Technology Architecture = 3, for a total of 36 entities. (pp. 65, 74, 93, 96, 99, 108)

Q: What is the key definitional difference between a "capability" (General Entity) and a "business capability" (Business Architecture entity)?

A: A business capability adds an actionable component — it must be exercised "to achieve a specific purpose or outcome" — whereas a plain capability is simply an ability without a stated purpose, making the business capability inherently more meaningful/valuable. (p. 73)

Q: What are the three types of measures described in the chapter?

A: Key Performance Indicators (KPIs) — outcome-focused, tied to CSFs; Metrics — broader quantitative/qualitative measures of process efficiency and operations; and Target Values — desired performance levels used as a reference point. (p. 78)

Q: In which ADM phases does gap analysis (producing the "gap" entity) occur?

A: Phases B, C, and D of the Architecture Development Method. (p. 70)

Q: What is the essential distinction between a value stream and a business process?

A: A value stream represents overall value-creating behavior from the stakeholder's perspective (non-linear, reflecting the business model) and delivers intangible value; a business process describes the time-ordered sequence of activities for an individual case (reflecting the operating model) and may produce a tangible end product. (pp. 80-82)

Q: How does the TOGAF Standard distinguish between the entities "function" and "process"?

A: Function is broader and more abstract, defining what activities are necessary to support organizational goals; process is narrower, defining how things get done step by step. (p. 84)

Q: What key distinction does TOGAF draw between "actor" and "organization unit"?

A: An actor can represent an entire organization but never a department within it; representing a department/sub-unit is exclusive to the organization unit entity. (p. 85)

Q: Which entity does the author recommend adding to the Enterprise Metamodel that is not formally present in the base TOGAF model, and why?

A: Stakeholder. TOGAF's organization unit entity is effectively used to represent stakeholders when related to goals, objectives, and measures, but TOGAF never explicitly names a stakeholder entity; ArchiMate does recognize a Stakeholder element for this purpose, so adding it improves TOGAF/ArchiMate integration. (pp. 104-105)

Q: Which entity acts as the structural link between Business Architecture and the Data/Application/Technology Architectures?

A: Business Service. (p. 91)

Q: How does ArchiMate's treatment of driver, goal, objective, and measure differ from TOGAF's?

A: ArchiMate classifies these as "motivation elements," a distinct category; TOGAF has no separate motivational/strategic category and instead places them all under the single heading of Business Architecture, since TOGAF only organizes entities into the four architecture domains (Business, Data, Application, Technology). (p. 108)

Q: What are the five steps of the process for tailoring the Enterprise Metamodel to an organization?

A: (1) Understand the Enterprise Metamodel, (2) Customize the Metamodel, (3) Integrate the Customized Metamodel (into the Content Framework), (4) Align Deliverables (with ADM phases), (5) Continuously Evolve. (pp. 106, 109-110)

Q: What does SMART stand for in the definition of "Objective"?

A: Specific, Measurable, Achievable, Realistic, and Timebound. (p. 77)